

# Tianhao Fei

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## Personal Information

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- **Date of Birth:** March 6, 2002
- **Marital Status:** Single
- **Gender:** Male
- **Home:** Anyue County, Ziyang City, Sichuan Province, China

## Education

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**Master, University of Science and Technology Beijing**

September 2024 – June 2027

*School of Automation and Electrical Engineering, Control Engineering*

- Score: 90.1/100; GPA: 4.0/4.0.
- Supervisor: **Yongliang Yang**

**Bachelor, University of Science and Technology Beijing**

September 2020 – June 2024

*School of Automation and Electrical Engineering, Automation*

## Research Interests

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Robotics, Reinforcement Learning, Adaptive Dynamic Programming, Optimal Control, Game Theory, Autonomous Driving, Unmanned Systems.

## Research Experience

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### 1. Adaptive Backstepping Control for Constrained Uncertain Nonlinear Systems

- **Adaptive control under multiple uncertainties:** Developed backstepping-based neural control schemes for nonlinear constrained systems with unknown parameters, stochastic disturbances, and unknown control directions.
- **Distributed synchronization and robust tracking:** Designed command-filtered and compensation-based controllers for heterogeneous switched multi-agent systems and time-delay dynamics under finite-energy disturbances.
- **Stability and performance guarantees:** Established boundedness, constraint satisfaction, bipartite synchronization, and robust  $H_\infty$  tracking performance through Lyapunov-type, and zero-sum game formulations.

### 2. Projected Self-Triggered Reinforcement Learning for Differential Games

- **Self-triggered learning:** Proposed communication-efficient learning frameworks for zero-sum and nonzero-sum differential games, avoiding continuous state monitoring while preserving closed-loop stability.
- **Projected safe adaptation:** Incorporated projection operators into Actor–Critic and Actor–Critic–Disturbance learning laws to prevent parameter drift and keep adaptive weights within prescribed compact sets.
- **Resource-aware Nash approximation:** Combined intermittent feedback, fuzzy logic, and experience replay to approximate Nash policies with reduced sampling, bounded learning errors, and Zeno-free execution.

### 3. Data-Efficient Composite Actor–Critic Reinforcement Learning

- **Composite learning for game-theoretic problem:** Developed Actor–Critic learning algorithm for multi-player nonzero-sum differential games by integrating instantaneous Bellman errors with historical data.
- **Relaxed excitation requirements:** Replaced restrictive persistence of excitation condition with finite/sufficient excitation-based learning, improving the practicality of online adaptive dynamic programming.
- **Improved convergence and stability:** Established convergence of critic/actor weights and closed-loop stability while enabling approximate Nash equilibrium learning for nonlinear multi-agent decision systems.

## Publications

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### Journal:

1. T. Fei, *et. al*, “[Game-Theoretic Fuzzy Reinforcement Learning via Self-Triggered Mechanism](#)”. IEEE Transactions on Fuzzy Systems, early access, 2026, doi: 10.1109/TFUZZ.2026.3706062. **(JCR Q1)**
2. T. Fei, *et. al*, “[Self-Triggered  \$H\_\infty\$  Control Using Projected Actor-Critic-Disturbance Learning](#)”. IEEE Transactions on Emerging Topics In Computational Intelligence, early access, 2026, doi: 10.1109/TETCI.2026.3683743. **(JCR Q1)**
3. T. Fei, *et. al*, “[Adaptive Neural Design for Permanent Magnet Synchronous Motor with Asymmetric Constraints and Input Dead-Zone](#),” Neurocomputing, vol. 640, p. 130294. **(JCR Q1)**
4. T. Fei, *et. al*, “[Command Filtered Adaptive Bipartite Synchronization for Heterogeneous Switched Multi-Agent Systems with State Delays and Input Nonlinearity](#),” Journal of the Franklin Institute, vol. 363, no. 2, p. 108395. **(JCR Q1)**
5. T. Fei, *et. al*, “[Robust Adaptive  \$H\_\infty\$  Tracking Control for Nonlinear Strict-Feedback System via Actor-Critic-Disturbance Learning: A Zero-Sum Game-Based Approach](#),” International Journal of Robust Nonlinear Control, vol. 36, no. 8, pp. 4517-4536. **(JCR Q1)**

### Conference:

7. T. Fei, *et. al*, “[Self-triggered Sampling Design For Continuous-Time Linear Dynamical System](#),” 2025 China Automation Congress (CAC), Harbin, China, 2025, pp. 3164-3169.
8. T. Fei, Y. Yang\*, “[Self-Triggered Nash Equilibrium Seeking for Multi-Player Linear Quadratic Differential Games](#),” 2026 Chinese Control and Decision Conference (CCDC), Nanjing, China, 2026. **Accepted, Oral presentation.**
9. T. Fei, *et. al*, “[Communication-Efficient Reinforcement Learning for Linear Quadratic Differential Zero-Sum Games](#),” 2026 International Annual Conference on Complex Systems and Intelligent Science (CSIS-IAC), Hefei, China, 2026. **Accepted, Best Paper Award.**

### Patent:

10. Y. Yang, T. Fei, *et. al*, “[Self-Triggered Robust Reinforcement Learning Method and Apparatus for Zero-Sum Games of Unmanned Surface Vehicles](#)”. Publication No. CN122151484A.
11. Y. Yang, T. Fei, *et. al*. “[Resilient Control for Power Buffers in DC Microgrids Against Denial-of-Service Attacks](#)”. Publication No. CN121566407A.
12. Y. Yang, Z. Shan, T. Fei, *et al*. “[Composite Game-Theoretic Reinforcement Learning for Quadrotor UAV Swarms](#)”. Publication No. CN121879395B.
13. Y. Yang, Z. Shan, T. Fei, Q. Li. “[Secure Control Method and Device for Power Buffers in DC Microgrids Against FDI Attacks](#)”. Publication No. CN121440520A.

## Honors and Awards

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- **2026 China Moutai “Pillars of the Nation” Elite Talent Award**, [China Youth Development Foundation](#) & [Kweichow Moutai Group](#). (150 recipients nationwide)
- **2026 Best Paper Award**, 2026 International Annual Conference on Complex Systems and Intelligent Science.
- **2026 Commemorative Certificate for Research Achievement in Invention Patent**, “Guohong Zhongjing Cup”, School of Automation and Electrical Engineering, University of Science and Technology Beijing (USTB).
- **2025 Excellent Merit Master Student**, USTB. (TOP 5%)
- **2025 Inspur “Langchao” Enterprise Scholarship**, [Inspur Group Co., Ltd.](#) (16 recipients university-wide)
- **2025 First-Class Academic Scholarship for Master Students**, USTB. (TOP 7%)
- **2023 Second Prize, Mathematical Competition** of USTB.
- **2022 Social Practice Gold Award**, USTB.

## Skills

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**Languages** Mandarin Chinese (Native), **CET-6**.

**Coding** Matlab, Python,  $\LaTeX$ ...

**Misc.** Academic research, teaching, training, consultation,  $\LaTeX$  typesetting and publishing.